CS383 · SOFTWARE ENGINEERING

Midterm 471 Solutions

20 marks

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01 Solution Q1: MCQ

#	Answer	Source
1	**Security/trust vulnerabilities and internet dependence**	The SaaS slide, Drawbacks column. Lecture 2.
2	**Application type, customer requirements, team background, and tools**	The Application types slide. Lecture 1.
3	**Inflexible to changing requirements and late problem discovery**	The waterfall model slide, Weaknesses. Lecture 3.
4	**Prototyping & incremental delivery**	The Coping with change slide. Lecture 3.

02 Solution Q2: User / System and FR / NFR

Requirement	User vs. System	FR vs. NFR
"The system should be easy to use for first-year students."	**User**	**NFR**
"The application shall allow password reset using a verified email."	**System**	**FR**
"The report generation shall be fast."	**User** *	**NFR**
"The interface will display student GPA to two decimal places."	**System**	**FR**

- User = a general statement in natural language. System = a specific, detailed statement fit for developers.
- FR = a service the system provides (allow, display). NFR = a quality attribute (easy to use, fast).
- The word shall does not make a statement functional. "shall be fast" is a quality attribute, so it is an NFR.

* The statement is general and has no measurable number, so it is User.

03 Solution Q3: sprint calculations

1) Sprint Velocity $= \frac{22 + 28 + 26 + 24}{4} = \frac{100}{4} = 25$

2) Focus Factor $= \frac{25}{6 \times 8} = \frac{25}{48} = 0.52$

3) Working days = 3 weeks × 5 days = 15 days

Capacity = 0.52 × (6 × 8 × 15) = 0.52 × 720 ≈ 375 hours

4) 30 > 25 → Prudent

- The focus factor here is `0.52`, which is below the optimal range (0.6 to 0.8).
- In part 3 we drop two days from each week (the weekend), so a week has 5 working days.
- In part 4: the backlog (30) is slightly larger than the velocity (25), so the plan is Prudent: possible with caution, and it is better to reduce the backlog.

Case	Verdict
Backlog ≈ Velocity	**Feasible**
Backlog > Velocity (slightly)	**Prudent**
Backlog >> Velocity	**Not feasible**

04 Solution Q4: process model

Project	Answer	Evidence in the text
1 · Smart-City Sensor Hub	**Component-Based**	"Must integrate COTS gateways and vendor SDKs": the system is built from ready-made components.
2 · Banking Regulatory Reporting	**Plan-Driven**	Limited stakeholders, low volatility, 30 people, documentation and sign-offs.
3 · Campus Social-Learning MVP	**Agile**	Daily feedback, high volatility, 6 co-located developers, a beta in 4 to 6 weeks.
4 · Medical Device Companion App	**Plan-Driven**	Compliance with IEC 62304 and ISO 14971 needs full life-cycle documentation.

How to decide

Look for three signals. COTS or a ready SDK as the main constraint points to Component-Based. Daily feedback, frequent change and a small team point to Agile. Regulation, documentation, sign-offs and a large team point to Plan-Driven. Project 4 has an SDK, but compliance dominates, so the answer is Plan-Driven.

05 Solution Q5: Product / Organization / External

#	Answer	Reason
1	**Product**	The performance of the product itself (response time).
2	**Organization**	A constraint from the company: a set language and platform.
3	**External**	A law from an outside body (HIPAA).
4	**Product**	The product's run-time behaviour.
5	**Product**	It concerns the product itself. Neither the company nor an outside body imposed it.
6	**External**	An external standard (ISO/IEC) for interoperating with other systems.

The rule: Product = a property of the product itself. Organization = a policy or procedure of the customer's or developer's company. External = a law, a standard or a body from outside.